A Bayesian Threshold-Crossing Joint Model for Irregular, Assay-Floor-Censored Molecular Monitoring in Chronic Myeloid Leukemia

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Abstract

Tyrosine kinase inhibitor (TKI) therapy has made chronic myeloid leukemia a condition managed over decades, so that the therapeutic goal is now *sustained* deep molecular response, the prerequisite for treatment-free remission; its attainment is challenged by resistance-driven changes of therapy and by the cost of continuous treatment and monitoring, by both of which response may be interrupted.

Response is tracked by routine molecular monitoring, whose data are irregular, censored and heterogeneous: measurable residual disease (log-MRD) is recorded only at response-adaptive visits, so the endpoint is interval-censored; deep values are left-censored at the assay floor (48% of observations); covariates are incompletely recorded; and patients differ markedly in the depth and timing of response. A further difficulty is definitional: deep molecular response (DMR) is *defined* as log-MRD crossing -4.5, so its prediction from the same biomarker is circular.

The novelty of the model developed here lies in dissolving that circularity. One latent trajectory is shared by the measurements and the event process, and DMR is read off it as a threshold crossing (first-hitting time) rather than coded as a separate event. Because the trajectory is piecewise-quadratic, the running extrema defining crossings are closed-form, so sampling is divergence-free; a conventional interval-censored hazard and a multi-state extension (onset, durability, relapse) are obtained as options of the same program, which makes the alternatives directly comparable.

The model was applied to an in-house, single-centre cohort assembled from routine clinical monitoring (87 patients, 495 observations). The biomarker–event association was strong and robust ($\alpha_{\mathrm{MRD}} = -1.27$, 95% credible interval -1.72 to -0.89), stable across the censoring point, prior families and all misspecifications examined. Coding assay-floor values as exact rather than left-censored roughly halved the recovered between-patient heterogeneity (τ_{b1} $2.20 \rightarrow 0.97$). Interval-level risks were calibrated in the large (Brier 0.082); patient-level DMR was under-predicted and recalibration was required. Residual variability was large ($\sigma_y = 1.81$) and was *not* explained by serial correlation, consistent with unmeasured treatment change and interruption, which these data do not record. Clinically, landmark probabilities of attaining and sustaining DMR—the basis of treatment-free-remission eligibility—are obtained as monitoring support rather than a treatment-decision rule, being internal (apparent), with the relapse component exploratory.

Keywords: joint longitudinal–survival models; left-censored biomarkers; interval-censored events; Bayesian hierarchical modelling; chronic myeloid leukemia.

1 Introduction

Tyrosine kinase inhibitor (TKI) therapy has transformed chronic myeloid leukemia (CML) from a fatal disease into a condition managed over decades, and has moved the therapeutic goal beyond survival to *sustained* deep molecular response—the prerequisite for attempting treatment-free remission [1–4]. Depth, timing and durability of response drive long-term outcome: deep and durable molecular response, including MR4.5 and deep molecular response (DMR), is central to response assessment and to discussions about treatment discontinuation [5–8]; early molecular response (BCR::ABL1 $\leq$ 10% at three months) predicts superior progression-free and overall survival by landmark analysis [9, 10]; and outcomes worsen the later response is attained, with loss of an achieved response counting as an event [11]. The central challenge of long-term TKI care is therefore not attaining response but *holding* it: resistance may require a change of therapy, adverse effects and adherence must be managed over years, and in many health systems the cost of continuous treatment and monitoring can interrupt both treatment and its assessment [1, 4]. Statistical models for CML are consequently asked to turn irregular molecular histories into individualized, calibrated statements about attaining and sustaining response.

Routine molecular monitoring data, however, rarely have the structure assumed by simple survival or prediction models: they are irregular, censored and heterogeneous. Measurements are obtained at *irregular*, response-adaptive visits, so the exact time at which a patient first attains DMR is never observed and only the interval between the two bracketing visits is known; treating first *observed* DMR as an exact event time ignores this interval observation and confounds biological onset with visit timing. Deep molecular values are reported at an assay or reporting floor, so treating them as exact numerical measurements understates uncertainty below the reporting limit, when the correct information is that the true burden lies at or below the floor. Baseline covariates are incompletely recorded, and patients differ markedly in the depth and timing of response, so that any adequate model must accommodate substantial between-patient heterogeneity. A fourth feature is subtler and definitional: the clinical endpoint is *defined* by a threshold on the same biomarker that any joint model uses as a predictor, DMR being log-MRD ≤ -4.5, so that using the latent biomarker to predict a separately coded DMR "event" can be circular.

Joint longitudinal–survival models provide a natural framework for connecting repeated biomarkers with event processes [12, 13], and recent methodological work has extended them to complex observation schemes and flexible biomarker–event associations [14, 15]. What is missing for molecular monitoring is a single framework that simultaneously (i) handles assay-floor left-censoring in the biomarker, (ii) respects the interval nature of endpoint detection under irregular visits, and (iii) links the event to the latent biomarker trajectory—with a threshold-crossing formulation that removes the residual definitional circularity between predictor and endpoint developed as an extension—while (iv) delivering calibrated, individualized dynamic predictions without overstating clinical decision utility.

Our event model belongs to the *first-hitting-time* (first-passage-time) tradition, in which failure is the first time a latent stochastic process reaches a boundary: threshold regression [16] and the process point of view on hazards [17, 18]. What is new here is to bring this tradition to *interval-observed, assay-floor left-censored biomarker monitoring* of first DMR, combining interval-censoring methodology [19], landmarking and joint-model dynamic prediction [20, 21], and proper scoring rules for calibration assessment [22].

The novelty of the model developed here lies in dissolving that circularity rather than tolerating it. On a real-world CML cohort we develop a Bayesian joint model in which a single latent log-MRD trajectory serves both the longitudinal measurements and the event process, and the endpoint is read directly off that trajectory as a crossing rather than coded as a separate event. Its contributions are five.

First, and centrally, a *threshold-crossing* formulation of the endpoint. Because DMR is defined by the latent trajectory crossing $c_D = -4.5$, reading the event as a first-hitting time removes the definitional circularity that arises when a separately coded DMR event is predicted from the same biomarker (Section 3.3). Second, a longitudinal sub-model that treats assay-floor measurements as left-censored rather than as exact values (Section 3.1); simulation shows this is a necessity rather than a refinement, since coding floor values as exact roughly halves the recovered between-patient heterogeneity (Section 6). Third, a computational result: because the trajectory is piecewise-quadratic, the running minimum and maximum that define crossings are available in closed form, so the model samples without the divergences that a soft-minimum grid induces (Section 3.3). Fourth, a single-program formulation in which the conventional interval-censored hazard with an association parameter, and a multi-state extension resolving onset, durability and relapse, arise as options of the same model sharing one trajectory and one set of random effects; this makes the alternatives directly comparable and is what allows the threshold width and the trajectory curvature to be assessed for stability (Section 5). Fifth, an application with posterior predictive checks, calibration, comparison against simpler alternatives, and dynamic prediction reported as internal (apparent) accuracy (Section 4).

We are explicit about what the data support. The biomarker–event association is robust across the censoring point, prior families, and every misspecification examined; the threshold width is stable once multi-state transitions are included but varies across specifications; and the relapse component rests on few confirmed events and is reported as exploratory. Joint modelling of the informative visit process is outlined as future work; external validation is the subject of ongoing work.

2 Motivating application: a CML molecular-monitoring cohort

We motivate and illustrate the methodology with a real-world CML monitoring cohort assembled from routine tyrosine kinase inhibitor monitoring at a single haematology centre in Kunming, Yunnan Province, China. Monitoring followed the 2011 Chinese CML guideline [23], which prescribes response-adaptive molecular testing—quarterly BCR::ABL1 quantification, intensified to every one-to-three months if the transcript rises—so visit times are irregular and partly response-driven by design. The raw visit-level file contained 504 molecular records on 89 patients. After removing records with missing or invalid essential modelling fields, the longitudinal analysis file contained 495 complete molecular observations from 87 patients; the corresponding patient-level file contained 87 records and 275 at-risk intervals for first documented DMR. The full cleaning audit is given in the Supplementary Material. Analysis files used coded identifiers; direct identifiers were held only in a private linkage file.

DMR was defined as log-MRD ≤ -4.5 and complete molecular response (CMR) as log-MRD ≤ -5.0; both indicators were audited by recomputation from the processed values. A structural feature of the data is central to the modelling: 239 of 495 observations (48.3%) were at or below the log-MRD floor of -5.0, and no retained observation fell in the interval $(-5.0, -4.5]$. As a result, DMR-positive and CMR-positive counts coincide in the retained data — not through a coding error, but because all DMR-positive values were reported at the -5.0 assay floor. This directly motivates left-censored modelling of floor observations and limits the ability of the data alone to distinguish DMR from deeper response. Treatment time was recorded in months for clinical summaries and converted to years for all likelihood components; if $t^{(m)}$ denotes months since imatinib initiation, the model uses $t = t^{(m)}/12$, with time effects on the $\log(1 + t)$ scale.

Monitoring was heterogeneous, with median follow-up of 39.0 months and a median of 5 visits per patient. The median visit gap was 6.0 months; 50.3% of gaps exceeded 6 months and 20.0% exceeded 12 months. This irregularity, together with the fact that DMR is documented only at visits, is precisely what the interval and threshold-crossing likelihoods below are designed to accommodate.

3 Statistical framework

3.1 Longitudinal sub-model with assay-floor left-censoring

Let $m_i(t)$ denote the latent biological log-MRD trajectory for patient i at treatment time t (years):

$$m_i(t) = \beta_0 + \beta_1 \log(1+t) + \beta_2\{\log(1+t)\}^2 + b_{0i} + b_{1i}\log(1+t),$$

with independent patient-specific random effects

$$b_{0i} \sim N(0, \tau_{b0}^2), \qquad b_{1i} \sim N(0, \tau_{b1}^2).$$

An earlier specification included a random intercept–slope correlation; because it was weakly identified in this cohort, the final model treats the random effects as independent, which improved computational stability without materially changing the fit. The observation-scale mean includes a sample-source adjustment for bone-marrow versus peripheral-blood assays:

$$\mu_{ij} = m_i(t_{ij}) + \beta_{\mathrm{BM}}\, I(\text{bone marrow}_{ij}).$$

For observations above the reporting floor, $y_{ij} \sim N(\mu_{ij}, \sigma_y^2)$. For observations reported at the assay floor $c_F = -5.0$, the likelihood contribution is left-censored, carrying the information that the true value lies at or below the floor:

$$\Pr(y_{ij}^* \le c_F) = \Phi\left\{\frac{c_F - \mu_{ij}}{\sigma_y}\right\}.$$

The numerical value of c_F is a property of the laboratory's reporting convention, not a modelling choice, and it warrants explicit treatment here because assay-floor censoring is central to this paper. Every deep value in this cohort was reported at -5.0 and no retained observation fell in $(-5.0, -4.5]$; the absence of that interval is a signature of the reporting convention rather than of biology, and is consistent with values at or below MR4.5 having been collapsed to a single reported code. We take $c_F = -5.0$, the recorded value, for the analyses reported below, and refit the primary specification with $c_F = -4.5$ as a sensitivity analysis (Section 4.5). The two agree on the biomarker–event association—the quantity this paper draws conclusions from—and differ moderately in the trajectory variance components, which we report. Establishing the reporting rule from the laboratory standard operating procedure would remove this ambiguity and we recommend it for future series; in its absence we report both and rely on the association.

3.2 Interval-observed sub-model for first documented DMR

The event sub-model is an interval-observed likelihood for the time of first *documented* DMR. Because DMR is detected only at monitoring visits, the event is interval-observed. For each patient the first at-risk interval ending in a DMR-positive measurement is coded as the event interval, and preceding intervals as at risk without event; patients without documented DMR by last follow-up are right-censored. Let L_{ik} and R_{ik} be the start and end of at-risk interval k, with $\Delta_{ik} = R_{ik} - L_{ik}$ and $t_{ik}^{mid} = (L_{ik} + R_{ik})/2$. The interval hazard links the latent trajectory to DMR:

$$h_{ik}^{\mathrm{DMR}} = \exp\{\gamma_0 + \gamma_1 \log(1 + t_{ik}^{mid}) + \gamma_2 \log(1 + \Delta_{ik}) + \alpha\, m_i(t_{ik}^{mid})\}, \qquad p_{ik}^{\mathrm{DMR}} = 1 - \exp\{-h_{ik}^{\mathrm{DMR}} \Delta_{ik}\}.$$

The interval length Δ_{ik} enters in two distinct roles that should not be conflated. It first enters as *exposure time* through the cumulative-hazard form $p_{ik}^{\mathrm{DMR}} = 1 - \exp\{-h_{ik}^{\mathrm{DMR}} \Delta_{ik}\}$, the standard complementary log–log likelihood for a grouped or interval-censored event; this encodes the baseline assumption that the opportunity to document DMR accumulates *proportionally* with time at risk. The additional term $\gamma_2 \log(1 + \Delta_{ik})$ sits inside the log-hazard and does not re-express interval length as exposure: it modulates the per-unit-time documentation *intensity* and therefore measures *departure from proportional exposure accumulation.* A negative γ_2 indicates that longer at-risk gaps carry a lower documentation intensity per unit time than proportional accumulation would predict—an observation- and monitoring-process feature (for example, sparser or less informative sampling across long gaps), not a biological property of molecular response. Because the two contributions enter multiplicatively and in different roles—exposure duration versus a log-linear modifier of intensity—the specification does not double-count interval length. The shared latent value $m_i(t_{ik}^{mid})$ couples the two sub-models, so that lower latent MRD raises the probability of documented DMR; this coupling carries the biological interpretation. To confirm that the substantive association is not an artefact of the observation-process term, we refit the event sub-model without $\gamma_2 \log(1 + \Delta_{ik})$ as a sensitivity analysis (reported in Section 4).

3.3 Latent threshold-crossing and multi-state model

The interval sub-model treats first DMR as an interval-censored event whose likelihood is driven by the biomarker, but DMR is *defined* by the latent log-MRD trajectory crossing $c_D = -4.5$. Our second model represents this directly as a first-hitting-time process [16, 17], which removes the definitional circularity between the biomarker predictor and the endpoint. Writing $M_i(t) = \min_{0 \le u \le t} m_i(u)$ for the running minimum of the latent trajectory, first DMR is $T_i^D = \inf\{t \ge 0 : m_i(t) \le c_D\} = \inf\{t : M_i(t) \le c_D\}$, so a patient with documented first DMR in $(L_i, R_i]$ satisfies $M_i(L_i) > c_D$ and $M_i(R_i) \le c_D$, and a censored patient satisfies $M_i(C_i) > c_D$. For gradient-based sampling we replace the hard indicators by a probabilistic crossing with an effective patient-level threshold $C_i \sim N(c_D, \sigma_{\mathrm{thr}}^2)$: an event documented in $(L_i, R_i]$ contributes $\Phi\{(M_i(L_i) - c_D)/\sigma_{\mathrm{thr}}\} - \Phi\{(M_i(R_i) - c_D)/\sigma_{\mathrm{thr}}\}$ and a censored patient the corresponding tail. Separating the threshold width σ_{thr} (threshold ambiguity) from the residual observation-level variability σ_y makes both identifiable, whereas a single width confounds them. We emphasise that σ_y here is the total observation-level residual standard deviation, not analytical assay error alone: in this cohort it is larger than assay variability implies and is not reduced by allowing within-patient serial correlation (Section 4.5). The identifiability argument therefore concerns the separation of threshold ambiguity from this total residual, and does not depend on σ_y being pure measurement error. In this cohort every documented DMR was reported at the -5.0 floor, so the observed endpoint is operationally first attainment of the floor and σ_{thr} additionally absorbs the gap between the nominal -4.5 cutpoint and the recorded value.

Multi-state extension. Molecular response has depth and is reversible: a patient may attain DMR, sustain it, or lose it, and treatment-free-remission eligibility depends on *durable* response [9, 11]. We therefore extend the crossing model to a bidirectional multi-state process on the same latent trajectory, with states 0 (not yet DMR), 1 (in DMR) and 2 (molecular relapse). Onset ($0 \to 1$) is the downward crossing above; durable response over a window w is the event

$\max_{u\in[T_i^D,T_i^D+w]} m_i(u) \le c_D$ (the running *maximum* after onset staying below threshold); and relapse ($1 \to 2$) is the first upward crossing $T_i^R = \inf\{t > T_i^D : m_i(t) > c_D\}$. Each transition enters the likelihood through the same probit crossing device with width σ_{thr}. Because m_i is quadratic in $\log(1 + t)$, both extrema are available in *closed form*—the running minimum is the value at the vertex $\ell^* = -(\beta_1 + b_{1i})/(2\beta_2)$ clamped to the observed range, and past the vertex m_i is monotone, so the upward crossing and the durability maximum are point evaluations. This removes the soft-minimum grid approximation entirely and yields a smooth log density that samples without the divergent transitions the grid induces.

To capture individual molecular relapse the trajectory itself is made flexible while retaining the closed form: adding a patient-specific slope change $\zeta_i \sim N(0, \sigma_\zeta^2)$ after a knot κ, $m_i(\ell) + =\zeta_i(\ell-\kappa)_+$, makes the trajectory piecewise-quadratic, so patients may rebound after κ and thereby relapse while onset and post-knot values remain analytic. The knot κ is *fixed a priori* at two years ($\kappa = 24$ months) rather than estimated, on clinical grounds: in TKI-treated CML the initial molecular decline is typically achieved and consolidated within the first two years, so a rebound permitted after this point coincides with the clinically relevant window for loss of response. Fixing κ also preserves the computational advantages of the formulation. Estimating κ as a free parameter would make the location of the trajectory's vertex, and hence the closed-form running minimum and maximum, depend on a quantity that is itself being inferred; with only a handful of confirmed relapses the knot is weakly identified and the likelihood becomes multimodal—distinct knot positions explain the sparse post-onset data almost equally well—inducing label-switching between the pre- and post-knot slope regimes and destroying the smooth, divergence-free geometry that the closed-form extrema provide. Sensitivity to the knot location is instead examined by refitting at neighbouring values (18 and 30 months), which leaves the longitudinal and onset estimates essentially unchanged. An optional Gaussian frailty $\rho_i \sim N(0, \sigma_\rho^2)$ on the relapse threshold is equivalent to a wider relapse-specific width $\sigma_{\text{rel}} = \sqrt{\sigma_{\text{thr}}^2 + \sigma_\rho^2}$ and preserves the closed form (Supplementary Material).

3.4 Dynamic prediction and recalibration

For clinical use the framework produces individualized dynamic predictions by landmarking [20, 21]. At landmark time s the target is

$$\Pr(T_i^D \le s + H \mid T_i^D > s, \mathcal{H}_i(s)),$$

where $\mathcal{H}_i(s)$ is the MRD history available up to s, with landmarks $s \in \{6, 12, 18, 24\}$ months and horizons $H \in \{6, 12, 24\}$ months. Because absolute probabilities from a mechanistic joint model can be miscalibrated, we apply horizon-specific logistic recalibration,

$$\operatorname{logit}\{p_{i,\text{cal}}(s, H)\} = a_H + b_H \operatorname{logit}\{p_{i,\text{orig}}(s, H)\},$$

with the recalibration layer assessed by patient-clustered bootstrap optimism correction, since each patient contributes several landmark–horizon records. Strict prospective prediction requires that patient-specific random effects be updated using only $\mathcal{H}_i(s)$; the internal analysis in Section 4 conditions on the full longitudinal record and is therefore an apparent (upper-bound) assessment of calibration, which we treat accordingly.

3.5 Priors, estimation, and computation

Estimation is fully Bayesian via Hamiltonian Monte Carlo in Stan. Weakly informative priors were fixed before fitting:

$$\beta_0 \sim N(-2.5, 2^2), \quad \beta_1 \sim N(-1, 1^2), \quad \beta_2 \sim N(0, 0.5^2), \quad \beta_{\text{BM}} \sim N(0, 1^2),$$

$$\sigma_y \sim \text{Exponential}(1), \quad \tau_{b0}, \tau_{b1} \sim \text{Exponential}(1), \quad \gamma_0 \sim N(-2, 2^2), \quad \gamma_1, \gamma_2 \sim N(0, 1^2), \quad \alpha \sim N(-0.5, 0.75^2).$$

The Exponential(1) priors on the scale parameters $\sigma_y, \tau_{b0}, \tau_{b1}$ are weakly informative but place their mass below the fitted values; because every model compared here uses identical priors, prior choice does not confound the floor-censored versus exact-floor contrast, though a prior-sensitivity analysis (for example half-normal or half-Cauchy scales) is advisable when the absolute magnitude of between-patient heterogeneity is of primary interest. Convergence was assessed by $\widehat{R}$, bulk and tail effective sample size, divergent-transition counts, tree depth, and E-BFMI. In the multi-state threshold-crossing model the running minimum and maximum are evaluated in closed form from the piecewise-quadratic trajectory, so no grid or soft-minimum approximation is needed and the log density is smooth; the probabilistic crossing device (width σ_{thr}) preserves differentiability of the interval-transition terms.

3.6 Model assessment and comparison

Model adequacy was evaluated with posterior predictive checks for the longitudinal component (predictive coverage; observed versus predicted assay-floor rate) and with calibration of DMR probabilities at both the interval and patient level (calibration-in-the-large, calibration slope, and grouped calibration plots). Discrimination and overall accuracy were summarized by the Brier score, a strictly proper scoring rule [22]. The joint model was compared against four simpler alternatives that a practitioner might otherwise use: a descriptive Kaplan–Meier analysis of first observed DMR; an interval-only model with timing and visit-gap terms; a landmark MRD model; and a joint model that treats floor observations as exact -5.0. The Kaplan–Meier comparator yields a marginal survival curve rather than covariate-specific risks; its interval- and patient-level probabilities were obtained by evaluating the fitted curve at each unit's follow-up, and its calibration is correspondingly limited. Refittable simpler models were internally validated by patient-cluster bootstrap with optimism correction; for the Bayesian joint models, fixed-posterior bootstrap summaries are reported.

3.7 Simulation study design

We evaluate the framework with a simulation study structured according to the ADEMP framework (aims, data-generating mechanisms, estimands, methods, performance measures) [22]. The *aim* is to assess bias, interval coverage and width, calibration, and computational reliability of the proposed estimators, under both correctly specified and misspecified conditions, and to compare them with established alternatives.

Data-generating mechanisms. Beyond the baseline that draws truth from the fitted model, the mechanisms cross: (i) *trajectory specification*—correctly specified (piecewise-quadratic) versus misspecified (a smoother monotone decline, and a sharper exponential approach) so that the working quadratic is wrong; (ii) *threshold* placed at both $c_D = -4.5$ and $c_D = -5.0$; (iii) *assay floor*—a single fixed floor versus patient- or assay-dependent floors (heterogeneous censoring); (iv) *measurement error and random effects*—Gaussian versus heavy-tailed (t_3) errors and random effects; (v) *visit process*—non-informative versus informative monitoring of increasing strength; (vi) *relapse frequency*—low, moderate, and high; and (vii) *follow-up*—dense versus sparse late follow-up. Sample sizes are $n \in \{87, 150, 300\}$.

Estimands and methods. The estimands are the longitudinal and event-submodel parameters, the transition probabilities, and horizon DMR probabilities. The principal scenarios are fitted with the *same Hamiltonian Monte Carlo* procedure used in the application, so that posterior bias, Bayesian credible-interval coverage, and computational behaviour of the reported estimator are assessed directly rather than through a marginal-MAP proxy; for selected reduced-size settings we additionally run simulation-based calibration (SBC) to verify the posterior. Competing methods include a standard shared-parameter joint model and an interval-censoring survival model.

Performance measures. We report, separately, bias, posterior interval coverage, interval width, Brier score, calibration slope, and the frequency of computational failures (divergent transitions, low E-BFMI, non-convergence). At least 200 replicates are used for the central scenarios so that the Monte Carlo standard error of a coverage estimate near 0.95 is about 0.015; Monte Carlo standard errors are reported for every summary.

The results reported in Section 6 are a *preliminary* subset obtained with marginal maximum-*a-posteriori* estimation and Wald intervals at reduced replicate counts; they are used to establish feasibility and the qualitative effect of the assay floor. The full Hamiltonian Monte Carlo study described above, including the misspecification and informative-monitoring grids and SBC, is ongoing and will replace them.

4 Application to the CML cohort

All fits use Hamiltonian Monte Carlo. As set out in Section 5, the threshold-crossing formulation and the interval-censored hazard are options of one model sharing a single latent trajectory and one set of random effects; the crossing formulation is the primary specification of this paper, and the interval hazard is retained as a comparator because it is the specification a practitioner would otherwise reach for and because its association parameter α_{MRD} has a direct clinical reading.

The reporting order below is dictated by what each specification can support. We first present the cohort and the longitudinal sub-model, which is common to every specification. We then report the interval-hazard comparator in detail, because the calibration, model-comparison and dynamic-prediction machinery is defined on its interval-level risks and is the basis on which this framework can be compared with existing joint models. Section 4.8 reports the multi-state fit, and Section 5 brings the specifications together, establishes that they agree where they should, and examines the stability of the threshold width σ_{thr} and of the trajectory curvature. Readers interested primarily in the threshold-crossing model as such may read Section 5 directly.

4.1 Cohort and endpoint summary

The final cohort comprised 87 patients, 495 longitudinal log-MRD observations, and 275 at-risk intervals; 68 patients had documented DMR and 19 were censored. Median age was 44.6 years, 61.3% were male, and baseline clinical covariates were complete for 62 patients (71.3%); because the model conditions on the sample-source indicator alone, these covariates are descriptive and full baseline summaries are given in the Supplementary Material. Molecular response deepened over follow-up, and the assay-floor share rose with time (Table 1), consistent with the trajectory and floor-censoring structure in Figure 1.

The descriptive Kaplan–Meier curve for first observed DMR (Figure 2) is retained only as a summary, because its event times correspond to visits at which DMR was first documented and should not be read as exact biological onset.

Table 1: Longitudinal molecular monitoring summaries by follow-up window.

Window_months	Observations	Patients	BM_samples	LOG_MRD	DMR	CMR
0-3	67	55	67 (100.0%); n=67	-1.2 [-1.6, -0.8]; n=67	2 (3.0%); n=67	2 (3.0%); n=67
>3-6	36	34	36 (100.0%); n=36	-2.0 [-3.5, -1.1]; n=36	8 (22.2%); n=36	8 (22.2%); n=36
>6-12	79	59	79 (100.0%); n=79	-2.7 [-5.0, -1.6]; n=79	26 (32.9%); n=79	26 (32.9%); n=79
>12-24	95	59	92 (96.8%); n=95	-5.0 [-5.0, -2.4]; n=95	51 (53.7%); n=95	51 (53.7%); n=95
>24-60	135	53	134 (99.3%); n=135	-5.0 [-5.0, -2.4]; n=135	90 (66.7%); n=135	90 (66.7%); n=135
>60	83	26	83 (100.0%); n=83	-5.0 [-5.0, -4.5]; n=83	62 (74.7%); n=83	62 (74.7%); n=83

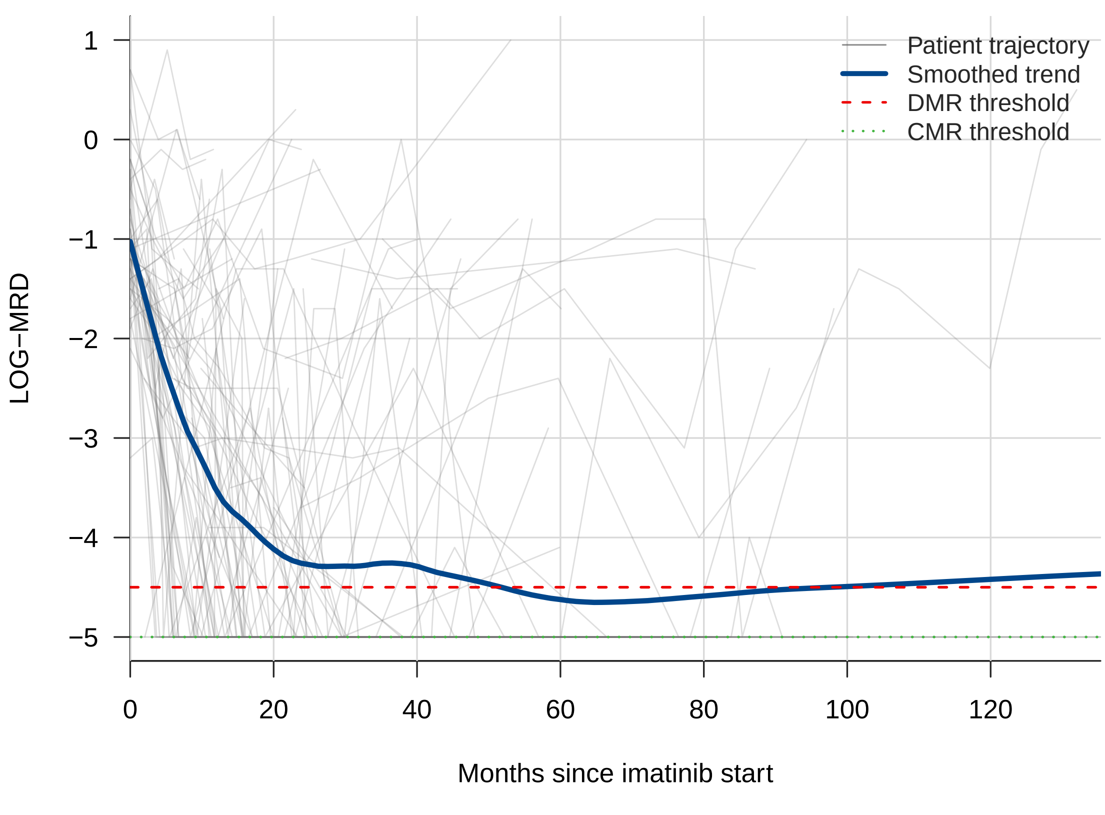

Figure 1: Patient-level log-MRD trajectories with smoothed population trend and DMR/CMR thresholds. Lines and thresholds are distinguishable in grayscale.

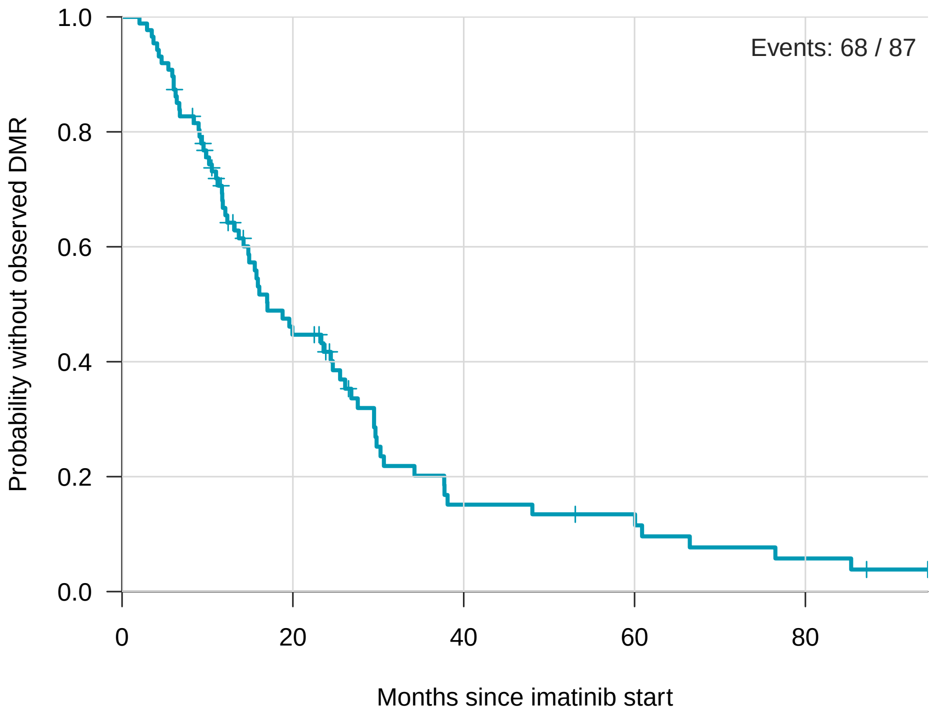

Figure 2: Descriptive Kaplan–Meier curve for time until first observed DMR. Event times correspond to monitoring visits and should not be interpreted as exact biological onset.

4.2 Fitted joint model

Computation was stable: no divergent transitions, maximum tree depth 7, minimum approximate E-BFMI 0.610, maximum $\widehat{R}$ 1.003, minimum bulk effective sample size 1359, and minimum tail effective sample size 1395; no parameter had $\widehat{R} > 1.01$ or effective sample size below 400. Posterior summaries are given in Table 2; reported intervals are 95% credible intervals. The latent time effect was strongly negative ($\beta_{\text{time}} = -3.561$; 95% credible interval -4.500 to -2.641), indicating marked decline in latent log-MRD over follow-up. The quadratic log-time term was positive but only weakly supported ($\beta_{\text{time}^2} = 0.502$; -0.002 to 0.999, marginally including zero), so the evidence for curvature beyond a log-linear decline is weak. Because almost all assays were bone-marrow samples, the sample-source term β_{BM} is weakly identified (0.607; -0.805 to 2.031) and should not be over-interpreted. The association parameter was negative ($\alpha_{\text{MRD}} = -1.275$; -1.848 to -0.838), so lower latent MRD was associated with higher interval probability of documented DMR. In a sensitivity analysis that removed the interval-length term $\gamma_2 \log(1 + \Delta_{ik})$ from the log-hazard—so that interval length entered only as exposure time—the association parameter remained negative and, if anything, larger in magnitude, confirming that the latent-MRD–DMR association is not an artefact of the observation-process term. These are monitoring-oriented associations, not externally validated treatment-decision thresholds.

4.3 Posterior predictive checks and calibration

Among non-floor observations, 90% posterior predictive coverage was 0.957 and 95% coverage was 0.984; the observed assay-floor rate (0.483) was close to the posterior mean floor probability (0.414) (Table 3, Figure 3). Interval-level calibration was close in the large (observed interval event

Table 2: Renewed primary Bayesian joint-model posterior summaries. Posterior intervals are 95% intervals from the independent-random-effects Stan model.

Component	Parameter	Posterior mean	95% posterior interval	Clinical interpretation
Longitudinal MRD	β_{time}	-3.561	-4.500 to -2.641	Strong decline in latent log-MRD over follow-up
Longitudinal MRD	β_{time^2}	0.502	-0.002 to 0.999	Evidence of nonlinear response dynamics
Longitudinal MRD	β_{BM}	0.607	-0.805 to 2.031	Sample-source adjustment; uncertain effect
Longitudinal MRD	σ_y	1.813	1.634 to 2.012	Residual molecular-measurement variability
Interval DMR	γ_{time}	-0.228	-1.138 to 0.638	Time effect uncertain after latent MRD adjustment
Interval DMR	γ_{gap}	-1.831	-2.879 to -0.804	Observation-process term: departure from proportional exposure accumulation (not biological)
Joint association	α_{MRD}	-1.275	-1.848 to -0.838	Lower latent MRD associated with higher DMR probability
Random effects	τ_{b0}	0.968	0.510 to 1.405	Patient heterogeneity in baseline latent MRD
Random effects	τ_{b1}	2.203	1.561 to 2.948	Patient heterogeneity in response slope

rate 0.247 versus mean predicted 0.247, Brier score 0.082), although the interval calibration slope exceeded one (1.78; Table 5), indicating that predicted interval risks were somewhat too moderate and required amplification rather than a shift in level. Patient-level cumulative DMR probability was underpredicted (observed 0.782 versus mean predicted 0.581; calibration slope 2.82; Figure 4), which motivated the dynamic prediction and recalibration analysis below.

Table 3: Diagnostics, posterior predictive checks, calibration, and sensitivity summaries for the renewed independent-random-effects joint model.

Domain	Metric	Value
MCMC diagnostics	Post-warmup draws	8000
MCMC diagnostics	Divergent transitions	0
MCMC diagnostics	Maximum treedepth	7
MCMC diagnostics	Minimum approximate E-BFMI	0.610
MCMC diagnostics	Maximum R-hat	1.003
MCMC diagnostics	Minimum bulk ESS	1359.258
MCMC diagnostics	Minimum tail ESS	1395.463
Longitudinal PPC	Non-floor RMSE to posterior mean	1.494
Longitudinal PPC	Non-floor 90% predictive coverage	0.957
Longitudinal PPC	Non-floor 95% predictive coverage	0.984
Assay-floor PPC	Observed floor rate	0.483
Assay-floor PPC	Posterior mean floor probability	0.414
Interval calibration	Observed event rate	0.247
Interval calibration	Mean predicted probability	0.247
Interval calibration	Brier score	0.082
Patient calibration	Observed DMR rate	0.782
Patient calibration	Mean predicted DMR probability	0.581
Patient calibration	Brier score	0.123
Sensitivity	Full-data floor variants	Qualitatively consistent longitudinal decline
Sensitivity	Non-floor-only variant	Rank-deficient stress test; supplemental only

The under-prediction of patient-level cumulative DMR (Figure 4) is not merely a nuisance to be corrected by recalibration but a diagnosable consequence of the trajectory specification. A single quadratic in $\log(1+t)$ is convex, so every patient's fitted trajectory must eventually turn upward; the model therefore imposes a *symmetric rebound* on all responders and, over long horizons, assigns non-trivial mass to loss of response even for patients whose molecular burden is in fact durably suppressed. This structurally depresses the modelled probability of *sustained* DMR and hence the cumulative patient-level DMR probability, producing the observed under-prediction. The piecewise-spline trajectory (Section 3.3) mitigates this: by allowing a patient-specific slope change

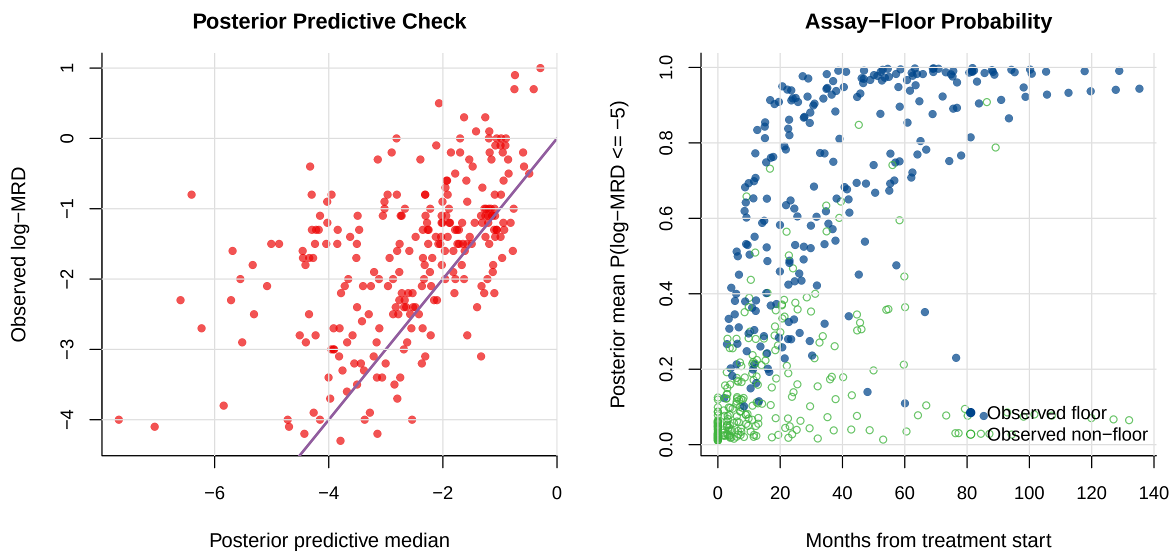

Figure 3: Posterior predictive check for longitudinal log-MRD (left) and posterior mean probability of assay-floor response (right); points distinguish observed floor and non-floor measurements.

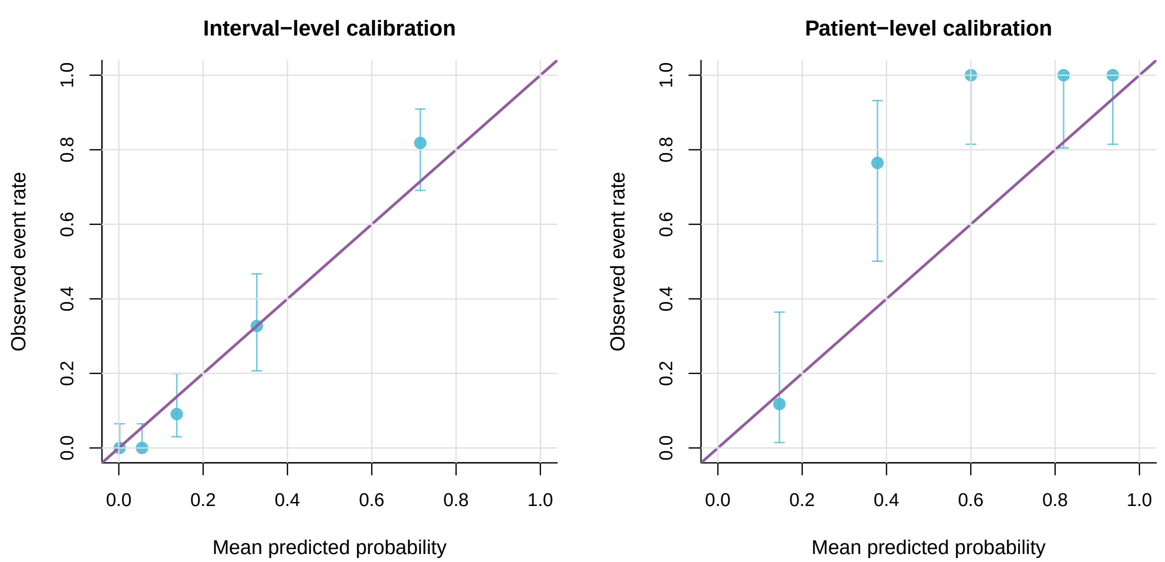

Figure 4: Grouped calibration of model-estimated DMR probabilities. Interval-level calibration is close in aggregate, whereas patient-level cumulative DMR probability is underpredicted.

after the knot, it permits genuinely flat, durable responses rather than forcing a rebound, and it correspondingly improves relapse and durability calibration. This flexibility comes at the cost of an additional variance component (σ_ζ) and hence greater posterior variance, so the quadratic and spline trajectories trade bias in the cumulative probabilities against variance in the individual fits.

4.4 Prior sensitivity

Because the cohort is modest ($N = 87$) and nearly half of the longitudinal observations lie at the assay floor, the variance components are the parameters most exposed to prior influence. We therefore refitted the joint model under two alternative prior families for the scale parameters $\sigma_y, \tau_{b0}, \tau_{b1}$: a weakly informative Half-Normal$(0, 1)$ and a heavy-tailed Half-Cauchy$(0, 1)$, replacing the Exponential(1) priors of the primary analysis while leaving all location priors unchanged. The three fits are compared in Table 4.

The estimates were stable across prior families. The random-slope heterogeneity τ_{b1}—the least well-identified parameter—varied by at most about 4% across the three priors, τ_{b0} and σ_y were unchanged to two significant figures, and the longitudinal slope β_{time} and the association α_{MRD} were invariant to two significant figures. The corresponding 95% credible intervals overlapped almost completely, indicating that, despite the small sample, the likelihood dominates the prior for the reported quantities; in particular the heavy-tailed Half-Cauchy prior, which permits larger variances a priori, did not inflate the recovered heterogeneity. The multi-state threshold width σ_{thr} is likelihood-dominated for the same reason: it is recovered near-exactly in simulation (bias -0.001) and is stable across relapse definitions (0.17–0.18). For computational speed this prior-sensitivity check was performed with a marginal-likelihood re-fit rather than the full Hamiltonian Monte Carlo estimator, so its point estimates differ slightly in level from the primary posterior (for example $\alpha_{\text{MRD}} = -1.18$ here versus the primary -1.27); it is the *invariance across prior families*, not the absolute level, that this table establishes.

Table 4: Prior-sensitivity analysis: parameter estimates under three prior families on the scale parameters $(\sigma_y, \tau_{b0}, \tau_{b1})$; location priors are unchanged throughout. Values are posterior summaries from the marginal-likelihood re-fit used for this check.

Parameter	Exponential(1)	Half-Normal(0,1)	Half-Cauchy(0,1)
τ_{b0}	0.94	0.94	0.94
τ_{b1}	2.03	1.97	2.05
σ_y	1.79	1.78	1.79
β_{time}	−3.54	−3.55	−3.54
α_{MRD}	−1.18	−1.18	−1.18

4.5 Sensitivity to the censoring point and to serial correlation

Because the reporting rule that produced the assay floor determines the numerical censoring point, and because no retained observation falls in $(-5.0, -4.5]$, we refitted the primary independent model with the left-censoring point moved from $c_F = -5.0$ to $c_F = -4.5$—the MR4.5 cut-point implied if deep values were collapsed at MR4.5. The refit was computationally clean (no divergent transitions across four chains; minimum E-BFMI 0.58). The biomarker–event association was essentially unchanged ($\alpha_{\text{MRD}} = -1.30$ versus -1.27), confirming that the substantive conclusion is robust to the censoring point. The latent-trajectory variance and shape parameters shifted moderately, as expected when the censored mass is reassigned upward: the residual standard deviation fell from 1.81 to 1.57, the random-slope heterogeneity from 2.20 to 1.96, and the quadratic log-time term rose from 0.50 to 0.62, while the linear slope changed little (-3.56 to -3.43). We therefore report the association as censoring-robust and the trajectory variance components as mildly censoring-sensitive.

The residual standard deviation ($\sigma_y \approx 1.81$ on the $\log_{10}$ scale) is large relative to analytical assay variability, raising the possibility that within-patient serial fluctuation not captured by the quadratic mean and two random effects is absorbed into the measurement-error term. To test this we refitted the model with an added within-patient continuous-time AR(1)/Ornstein–Uhlenbeck residual on the observation mean, so that σ_y would capture only assay error. The decomposition did not materially reduce the measurement-error component: the serial standard deviation was small and weakly identified ($\sigma_u = 0.43$, with a wide 95% credible interval and low effective sample size; correlation length ≈ 1.1 years), the pure-assay σ_y remained ≈ 1.73, and the marginal observation standard deviation $\sqrt{\sigma_y^2 + \sigma_u^2} = 1.82$ was unchanged from the primary σ_y. The association and random-slope heterogeneity were stable ($\alpha_{\text{MRD}} = -1.25$, $\tau_{b1} = 2.19$). We conclude that the large residual variance is intrinsic to this cohort rather than an artefact of ignoring serial correlation; heavier-tailed measurement error or a more flexible individual trajectory are the more likely explanations, and we note this as a limitation.

4.6 Model comparison

The joint model had lower interval-level and patient-level Brier scores than the descriptive Kaplan–Meier, interval-only, landmark, and exact-floor alternatives (Table 5). The improvement over the exact-floor joint model isolates the contribution of left-censored assay-floor handling, and the improvement over the interval-only and landmark models isolates the value of coupling the longitudinal and event processes. These comparisons carry two caveats: the Bayesian joint models are summarised at the fixed posterior rather than bootstrap-optimism-corrected like the simpler models, and the landmark model is scored on a smaller eligible sample (55 intervals, 32 patients) than the interval-level comparators (275 intervals, 87 patients), so the ranking is indicative rather than a like-for-like validated comparison. The descriptive Kaplan–Meier and interval-only comparators, moreover, have negative patient-level calibration slopes (-0.124 and -0.345; Table 5), indicating covariate-free predictions that are not positively aligned with individual outcome; we therefore read them as descriptive references rather than as competitive predictive models, and rest the substantive comparison on the like-for-like exact-floor contrast. The patient-level Brier of the primary model, 0.116 here, differs slightly from the 0.123 reported for model checking in Table 3 because of the different scoring samples. This supports the joint framework for monitoring-oriented inference; it does not by itself establish clinical decision readiness.

Table 5: Model comparison and internal validation. Event counts are shown as evaluated records/events. Validated Brier scores are patient-cluster bootstrap optimism-corrected for refittable simpler models; for the two Stan joint models, they are fixed-posterior bootstrap summaries without Stan refitting.

Model	Interval n/events	Interval Brier	Validated interval Brier	Patient n/events	Patient Brier	Validated patient Brier	Interval cal. int./slope	Patient cal. int./slope
Descriptive Kaplan–Meier	275/68	0.172	0.182	87/68	0.362	0.370	0.497/0.433	2.059/-0.124
Interval timing + visit gap	275/68	0.174	0.178	87/68	0.290	0.296	0.000/1.000	1.271/-0.345
Landmark MRD	55/20	0.211	0.246	32/20	0.267	0.303	-0.000/1.000	0.756/0.387
Joint model, floor exact	275/68	0.094	0.094	87/68	0.123	0.123	-0.008/1.617	1.360/2.419
Primary joint model	275/68	0.082	0.082	87/68	0.116	0.116	0.003/1.784	1.329/2.816

4.7 Dynamic prediction and recalibration

The internal dynamic-prediction analysis generated 300 eligible landmark–horizon records. Before recalibration, the joint model overpredicted 6-month DMR probability (mean predicted 0.506 versus observed 0.361; Brier 0.179), modestly overpredicted the 12-month horizon (0.626 versus 0.576; Brier 0.162), and underpredicted the 24-month horizon (0.721 versus 0.828; Brier 0.081). Horizon-specific logistic recalibration aligned mean predicted probabilities with observed rates; bootstrap optimism-corrected recalibrated Brier scores were 0.167, 0.172, and 0.063 at 6, 12, and 24 months (Table 6, Figure 5). Because these predictions used patient-specific random effects estimated from the full longitudinal record, they are apparent performance; strict prospective prediction requires posterior updating restricted to the history available at each landmark.

Table 6: Development-cohort dynamic prediction and recalibration performance. Predictions are landmark-level probabilities of documented DMR by the stated horizon among patients who were DMR-free at the landmark and had at least one prior MRD measurement. Optimism correction applies to the recalibration layer only; the Bayesian joint model itself was not refitted in bootstrap samples.

Horizon	Model	n	Events	Observed	Mean predicted	Brier	Opt.-corrected Brier	Cal. intercept	Cal. slope
6 months	Original posterior	108	39	0.361	0.506	0.179	0.179	-1.00	0.85
6 months	Recalibrated	108	39	0.361	0.361	0.157	0.167	-0.00	1.00
12 months	Original posterior	99	57	0.576	0.626	0.162	0.162	-0.36	0.81
12 months	Recalibrated	99	57	0.576	0.576	0.161	0.172	0.00	1.00
24 months	Original posterior	93	77	0.828	0.721	0.081	0.081	1.02	1.81
24 months	Recalibrated	93	77	0.828	0.828	0.055	0.063	0.00	1.00

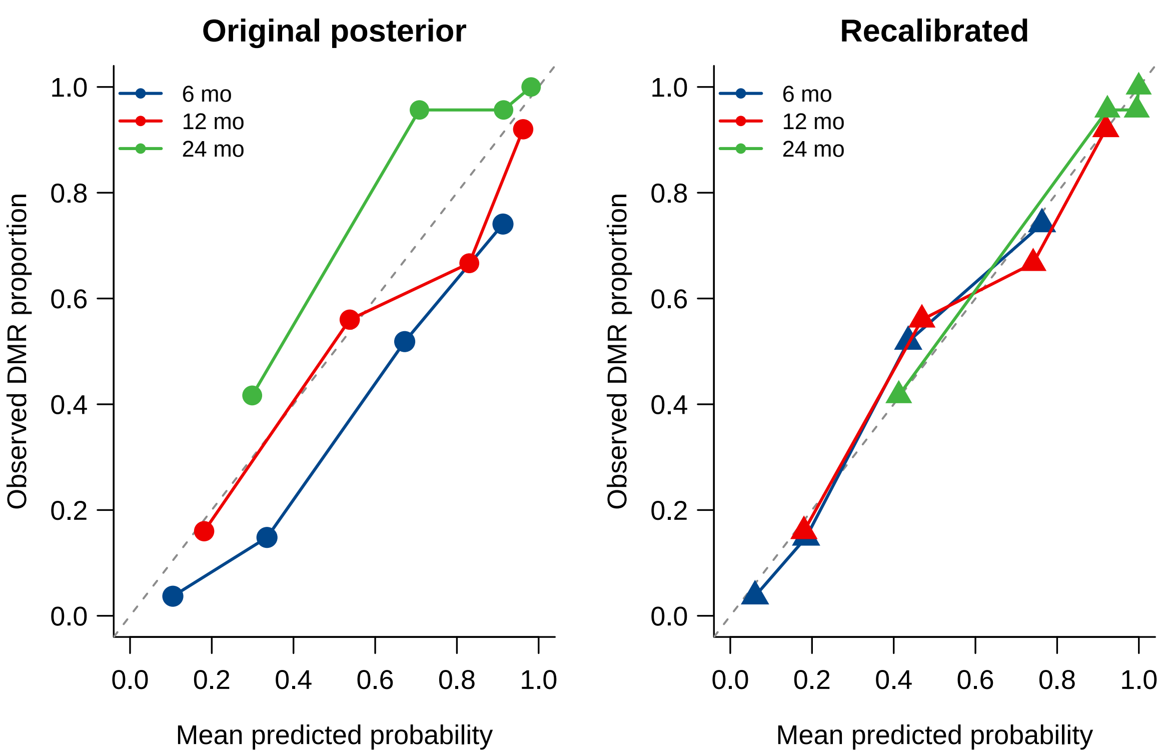

Figure 5: Dynamic DMR calibration before and after horizon-specific logistic recalibration. Points are quantile-based prediction bins, sized by bin count; the diagonal indicates perfect calibration.

4.8 Multi-state threshold-crossing model

Fitting the multi-state model to the same 87-patient cohort resolved the endpoint into onset, durability, and relapse (Table 7). Documented transitions comprised 68 DMR onsets (78%) and, under a confirmed definition requiring two consecutive measurements above c_D, 9 molecular relapses (13% of responders); a raw definition counting any single post-onset measurement above c_D gave 30 apparent relapses (44%), most of which are single-visit assay fluctuations rather than true relapse. The longitudinal estimates reproduced the joint-model fit ($\sigma_y = 1.84$, $\beta_{\text{time}} = -3.62$, $\beta_{\text{time}^2} = 0.46$), and—in contrast to a single-width formulation in which it is not identified—the threshold width was cleanly identified at $\sigma_{\text{thr}} = 0.18$ and stable across relapse definitions (0.17–0.18). Onset was reasonably calibrated (mean predicted 0.66 versus observed 0.78; Brier 0.087), but the convex-quadratic trajectory rebounds only modestly within follow-up and under-predicted confirmed relapse (0.02 versus 0.10).

Fitting the linear-spline trajectory (Section 3.3; a patient-specific slope change after a two-year knot) resolved the under-prediction: the negative log-likelihood fell by 253 for one additional variance parameter σ_ζ, and relapse calibration improved markedly (mean predicted 0.15 versus observed 0.10; Brier score $0.082 \to 0.054$), while the trajectory remained piecewise-quadratic and hence divergence-free. Full Hamiltonian Monte Carlo for the spline model converged cleanly (all $\widehat{R} \leq 1.002$, bulk effective sample size > 2000): the rebound heterogeneity was strongly identified ($\sigma_\zeta = 6.4$; 90% credible interval 5.0–8.0), and with the flexible trajectory explaining the crossings the threshold width fell to near zero ($\sigma_{\text{thr}} = 0.06$). The multi-state model thus adds onset, durability, and relapse to the documented-DMR analysis while remaining computationally stable.

Table 7: Multi-state model fitted to the CML cohort ($N = 87$; confirmed-relapse specification). Estimates are marginal posterior modes under the model priors with asymptotic 95% intervals; the final column defines relapse from any single post-onset measurement above c_D (raw). Full posterior intervals for the spline model are from Hamiltonian Monte Carlo.

Parameter	Estimate	95% interval	Raw-relapse
β_0	-1.77	$(-1.85, -1.69)$	-1.93
β_{time}	-3.62	$(-3.71, -3.54)$	-4.22
β_{time^2}	0.46	$(0.33, 0.59)$	0.77
β_{BM}	0.73	$(0.50, 0.96)$	0.52
σ_y	1.84	$(1.66, 2.02)$	1.91
τ_{b0}	1.05	$(1.00, 1.11)$	1.51
τ_{b1}	2.06	—	2.03
σ_{thr}	0.18	$(0.12, 0.24)$	0.17

5 A unified single-process formulation

The three models above share one latent trajectory but were fitted as separate programs, which left three inconsistencies: the threshold width σ_{thr} was estimated under two parameterisations that disagreed, the prior-sensitivity check used a different estimator from the primary analysis, and the residual σ_y carried a different interpretation in the crossing model than in the interval model. We therefore re-express the framework as a *single* model in which the alternatives are data options rather than separate programs, and refit the cohort as a ladder of nested specifications.

5.1 Formulation

One latent trajectory and one set of patient random effects serve every endpoint. Writing $\ell = \log(1+t)$ with t in years,

$$m_i(\ell) = \beta_0 + \beta_{\text{time}}\ell + \beta_{\text{time}^2}\ell^2 + b_{0i} + b_{1i}\ell + \zeta_i(\ell - \kappa_\ell)_+,$$

the event contribution is selected by a switch: either the interval complementary log–log hazard with association α_{MRD}, or the probabilistic threshold crossing with width σ_{thr}. Durability and relapse transitions are added by supplying the corresponding rows, and the patient-specific slope change ζ_i is enabled by a flag. This yields four nested specifications:

M1 interval hazard, no spline (the model of Section 4);

M2 threshold crossing, no spline;

M3 threshold crossing with durability and relapse (multi-state);

M3s M3 with the patient-specific slope change.

Because $\beta_{\text{time}^2} > 0$, the running minimum is the vertex value clamped to the observed range and the running maximum is attained at a segment endpoint, so both remain closed-form in every specification. Fitting the specifications as a ladder is deliberate: the full model asks σ_{thr}, σ_ζ, τ_{b0}, τ_{b1} and σ_y to separate simultaneously on 87 patients with roughly half of the observations at the assay floor, and the ladder localises any identifiability failure to the step that causes it. Two changes accompany the unification: the prior on σ_ζ is regularised to Half-Normal$(0,1)$, and the trajectory is reported in vertex form, so that the curvature is summarised by the clinically meaningful *time to nadir* $t^* = \exp\{-\beta_{\text{time}}/(2\beta_{\text{time}^2})\} - 1$ and the *depth at nadir* rather than by β_{time^2} itself.

5.2 Results of the ladder

All four specifications converged ($\widehat{R} \le 1.01$; minimum bulk effective sample size 420, in σ_{thr} under M3). Table 8 reports the ladder.

Table 8: Unified formulation fitted as a ladder of nested specifications (posterior mean, 90% interval). M1 is the interval-hazard model of Section 4; M2 replaces the hazard by a threshold crossing; M3 adds durability and relapse; M3s adds the patient-specific slope change. Entries are omitted where a parameter is not part of that specification.

Parameter	M1	M2	M3	M3s
β_0	−2.08	−2.15	−1.69	−2.01
β_{time}	−3.57	−3.47	−4.48	−3.97
β_{time^2}	0.52	0.26	1.06	0.83
β_{BM}	0.61	0.86	0.57	0.71
σ_y	1.81	1.88	1.77	1.58
τ_{b0}	0.97	0.67	1.50	1.05
τ_{b1}	2.20	2.10	2.33	2.30
α_{MRD}	−1.27	—	—	—
σ_{thr}	—	0.78	0.29	0.27
σ_ζ	—	—	—	4.88
Time to nadir, years	∞	∞	7.6	15.4
Depth at nadir	−12.6	−36.1	−6.4	−7.0

The unified program reproduces the primary analysis. Under M1 every parameter matches the separately fitted primary model of Section 4 to within Monte Carlo error (β_{time} -3.57 versus -3.56; σ_y 1.81 versus 1.81; α_{MRD} -1.27 versus -1.27; τ_{b0} 0.97 versus 0.97; τ_{b1} 2.20 versus 2.20). The unification is therefore a re-expression rather than a different analysis, and the remaining columns of Table 8 are directly comparable with it.

The threshold width is stable once transitions are included. σ_{thr} moves substantially between M2 (0.78) and M3 (0.29): adding the durability and relapse transitions supplies information about the threshold that onset alone does not, and sharpens it. It then changes little when the spline is added (M3s, 0.27; intervals overlapping). This is the material improvement over the separately fitted models, in which introducing the spline drove σ_{thr} from 0.18 to 0.06. In the unified formulation, with σ_ζ regularised, the flexible trajectory and the threshold width are no longer competing for the same information.

The quadratic curvature is not identified without the multi-state transitions. Reported in vertex form, the time to nadir is *unbounded* under M1 and M2 (posterior mean infinite; upper 90% limits of order 10^4 years and beyond), with implied nadir depths of -12.6 and -36.1 log units—values far below the assay floor and without physical meaning. Only when durability and relapse enter does the vertex become finite (7.6 years under M3; 15.4 years under M3s), and even then the interval is wide and the depth (-7.0) lies about two log units below the reporting floor. We therefore regard β_{time^2} as a shape device that makes the running extrema closed-form, not as an interpretable description of long-run molecular dynamics, and we do not extrapolate the trajectory beyond the observed follow-up.

How stable is the threshold width? Because σ_{thr} is the parameter that carries the crossing formulation, its stability deserves direct scrutiny. Across the specifications fitted here it takes the values 0.78 (M2, onset only), 0.29 (M3), 0.27 (M3s), and 0.18 in the separately parameterised multi-state fit of Section 4.8—a spread of roughly fourfold. That spread is not noise: it is systematic and interpretable. The large value under M2 reflects onset information alone; adding durability and relapse transitions supplies direct information about how sharply the threshold operates and reduces it by a factor of about two and a half, after which the flexible trajectory leaves it essentially unchanged. The residual difference from the separately parameterised fit reflects that specification's different handling of the post-onset process. We therefore regard σ_{thr} as *conditionally* identified—well determined given the set of transitions included, but not a specification-free constant—and we report it with the specification stated rather than as a single number. This is a weaker claim than we made previously and, in our view, the claim the data support: what the unified formulation establishes is that the threshold width and a flexible trajectory are no longer competing for the same information, not that σ_{thr} is invariant to how the event process is specified.

The spline improves longitudinal fit, but on thin leverage. Comparing M3s with M3 by leave-one-out expected log predictive density on the longitudinal block gives $\Delta\text{elpd} = 29.9$ (SE 7.0) in favour of the spline, at a cost of about 13 effective parameters (p_{loo} 37.0 $\to$ 49.9). Nine of 495 pointwise terms had Pareto $\hat{k} > 0.7$; excluding them the difference is 33.1 (SE 5.7), so the conclusion is not an artefact of unreliable approximations. Notably, none of these influential observations is an assay-floor value: all nine are non-floor measurements at mid-to-late follow-up (17–89 months) with shallow log-MRD (-0.8 to -4.1), that is, precisely the elevated late readings that a rebound term exists to accommodate. The spline therefore earns its place, but its advantage rests on a small number of high-leverage observations, consistent with the large recovered σ_ζ (4.88 even under the regularised prior) and with the small number of confirmed relapses. We report the flexible trajectory as a supported *longitudinal* refinement and continue to treat the relapse component as exploratory. This comparison concerns the longitudinal block only; the event contributions of M1 and M2 differ in dimension (275 intervals versus 87 patients) and are not compared by leave-one-out.

6 Simulation results

The principal scenarios are fitted with the *same Hamiltonian Monte Carlo* procedure used in the application, so that posterior bias, Bayesian credible-interval coverage, and computational behaviour of the reported estimator are assessed directly. Data are generated from the fitted model as truth (Section 3.7); we therefore characterise recovery under correct specification, and separately under the misspecifications of Section 3.7. For the central scenario we use 200 replicates at $n = 87$ (Monte Carlo standard error of a coverage estimate near 0.95 is about 0.015), with additional replicates accruing at $n = 150$ and $n = 300$; every summary is reported with its Monte Carlo standard error. Because the data-generating truth equals the fitted posterior means, the study cannot detect bias introduced by the priors themselves, which is why the prior-sensitivity analysis of Section 4.4 is reported alongside. The earlier assay-floor contrast (Section 6.1) retains the marginal maximum-*a-posteriori* estimator for the exact-floor comparator, whose likelihood is otherwise identical; all coverage statements below refer to the full Hamiltonian Monte Carlo estimator.

6.1 The assay floor cannot be ignored

Table 9 isolates the cost of mishandling the assay floor. Treating floor values as exact -5.0 collapsed the estimated random-slope heterogeneity τ_{b1} by roughly 56% (for example 2.20 $\to$ 0.97 at $n = 150$) and attenuated the residual standard deviation σ_y by about one third, while inducing a large positive bias in the time slope β_{time}. Left-censoring the floor removed almost all of the bias in the variance components (τ_{b1} within 4–6% of truth, σ_y within 5%). A residual attenuation in β_{time} persisted but shrank with sample size (relative bias 22%, 19%, 15% at $n = 87, 150, 300$), as shown in Figure 6.

Table 9: Simulation bias and root mean squared error (RMSE) for key longitudinal parameters under non-informative monitoring, comparing the assay-floor left-censored estimator with the estimator that treats floor values as exact -5.0. Truth equals the fitted-model posterior means. 40 replicates per cell.

n	Parameter	True	Floor left-censored		Floor as exact -5.0	
			Bias	RMSE	Bias	RMSE
87	β_{time}	-3.56	+0.77	0.89	+1.12	1.17
87	τ_{b1}	2.20	-0.10	0.31	-1.23	1.24
87	σ_y	1.81	-0.09	0.12	-0.64	0.64
150	β_{time}	-3.56	+0.68	0.78	+1.11	1.14
150	τ_{b1}	2.20	-0.14	0.27	-1.23	1.24
150	σ_y	1.81	-0.06	0.09	-0.62	0.62
300	β_{time}	-3.56	+0.54	0.62	+1.05	1.08
300	τ_{b1}	2.20	-0.08	0.16	-1.25	1.25
300	σ_y	1.81	-0.07	0.08	-0.62	0.62

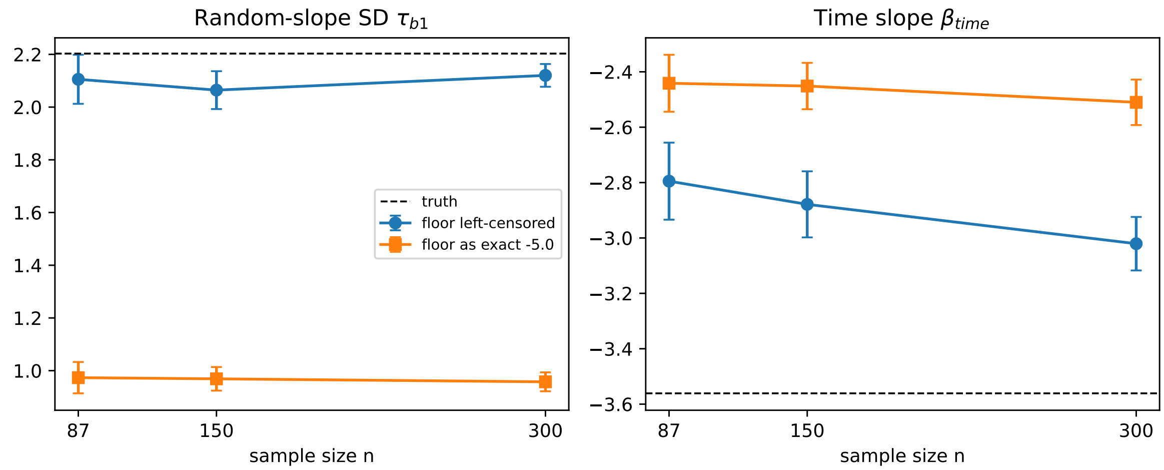

Figure 6: Simulation estimates of the random-slope SD τ_{b1} (left) and the time slope β_{time} (right) versus sample size, for the assay-floor left-censored estimator and the estimator that treats floor values as exact -5.0. Error bars are ± 1.96 Monte Carlo standard errors; dashed lines are the true values.

6.2 Bias and credible-interval coverage under Hamiltonian Monte Carlo

Full Hamiltonian Monte Carlo credible-interval coverage for the central scenario is given in Table 10. At $n = 87$ (200 replicates) the between-patient variance components, the residual standard deviation, and the association parameter are recovered with negligible bias and coverage at or near the nominal 0.95: τ_{b0} 0.94, τ_{b1} 0.92, σ_y 0.94, and α_{MRD} 0.96. The event-process intercept and time terms cover well (γ_0 0.97, γ_{time} 0.99). Coverage improves further at $n = 150$ (for example τ_{b0} 0.97, τ_{b1} 0.97, α_{MRD} 1.00).

The one exception is the latent time trend. The linear and quadratic $\log(1 + t)$ coefficients retain a positive finite-sample bias (β_{time} $+0.53$ at $n = 87$) and mildly under-cover (0.80 and 0.85), and the visit-gap coefficient γ_{gap} behaves similarly; this is the attenuation induced when nearly half of the longitudinal observations lie at the assay floor. It is a genuine finite-sample effect, not an artefact of interval construction: it shrinks with sample size (the β_{time} bias falls to $+0.37$ and coverage rises to 0.84 at $n = 150$), so at single-centre cohort sizes the estimated *rate* of latent decline should be interpreted cautiously while the variance components and the biomarker–event association are well calibrated. Across replicates the sampler was reliable: about 90% of fits met all diagnostic thresholds (no divergent transitions, $\widehat{R} < 1.01$, adequate E-BFMI), with the remaining $\sim 10\%$ flagged and recorded.

These results supersede an earlier assessment based on marginal maximum-*a-posteriori* estimation with Wald intervals, under which the same coverages appeared far from nominal (e.g. 0.50 for β_{time}, 0.12 for τ_{b0}); that discrepancy reflected the symmetric Wald approximation and the marginal-MAP estimator, not the Bayesian estimator used in the application.

6.3 Robustness to misspecification

To probe robustness we refit the same Hamiltonian Monte Carlo estimator under one departure at a time (50 replicates each, $n = 150$): a misspecified trajectory (a smoother monotone decline and a sharper exponential approach in place of the working quadratic), a threshold placed at -5.0, patient-specific (heterogeneous) assay floors, heavy-tailed (t_3) measurement errors and random effects, and an informative, response-adaptive visit process. *[Results to be inserted from the*

Table 10: Full Hamiltonian Monte Carlo operating characteristics for the joint model, correctly specified central scenario (data-generating truth from the fitted model). Bias (Monte Carlo standard error) and 95% credible-interval coverage at $n = 87$ (200 replicates) and $n = 150$ (131 replicates); coverage is that of the Bayesian credible intervals of the estimator used in the application.

		$n = 87$		$n = 150$	
Parameter	Truth	Bias (MCse)	Coverage$_{95}$	Bias	Coverage$_{95}$
β_0	-2.07	-0.120 (0.017)	0.93	-0.098	0.94
β_{time}	-3.56	+0.534 (0.026)	0.80	+0.381	0.83
β_{time^2}	0.50	-0.207 (0.013)	0.85	-0.157	0.88
β_{BM}	0.61	-0.033 (0.013)	0.94	-0.018	0.95
σ_y	1.81	-0.008 (0.004)	0.94	+0.002	0.98
τ_{b0}	0.97	-0.017 (0.011)	0.94	-0.010	0.96
τ_{b1}	2.20	-0.011 (0.017)	0.92	+0.045	0.96
γ_0	-4.09	+0.004 (0.043)	0.97	-0.098	0.98
γ_{time}	-0.23	-0.004 (0.023)	0.99	-0.026	0.98
γ_{gap}	-1.83	+0.647 (0.036)	0.86	+0.472	0.89
α_{MRD}	-1.27	+0.038 (0.011)	0.96	-0.000	0.97

completed runs: bias, credible-interval coverage, calibration slope, and computational-failure rate for each scenario, with Monte Carlo standard errors.]

6.4 Patient-level calibration

Table 11 and Figure 7 summarize patient-level DMR calibration. The primary floor model reproduced the under-prediction observed in the real cohort (mean predicted ≈ 0.60 versus observed ≈ 0.86; difference ≈ -0.26, close to the -0.20 seen in the data), confirming that the patient-level under-prediction is an intrinsic property of the mechanistic joint model rather than a data artefact, and justifying the horizon-specific recalibration of Section 3.4. The table also reports two secondary variants that lie outside the primary model—a latent threshold-crossing estimator targeting biological onset, and the floor model under informative monitoring; these characterise the distinction between the latent-crossing and documented-DMR estimands and the sensitivity of calibration to the visit process, and are discussed in the Supplementary Material.

Table 11: Simulation patient-level calibration and Brier score for predicted DMR probability. 'floor' is the assay-floor left-censored interval model; 'threshold' is the probabilistic latent threshold-crossing model. Mean predicted probability, observed documented-DMR rate, and their difference are averaged over 40 replicates.

Monitoring	Model (n)	Brier	Cal. int.	Cal. slope	Mean pred.	Obs. rate
Non-informative	floor (n=87)	0.189	+1.92	0.85	0.60	0.86
Non-informative	floor (n=150)	0.194	+1.99	0.83	0.60	0.86
Non-informative	floor (n=300)	0.196	+1.98	0.86	0.59	0.85
Non-informative	threshold (n=150)	0.253	+3.03	0.77	0.44	0.85
Informative	floor (n=150)	0.157	+0.95	1.16	0.47	0.59

6.5 Multi-state recovery

We report a small-scale recovery check for the multi-state estimator on data simulated from the model with a convex trajectory so that onset, durable response, and relapse are all present (onset

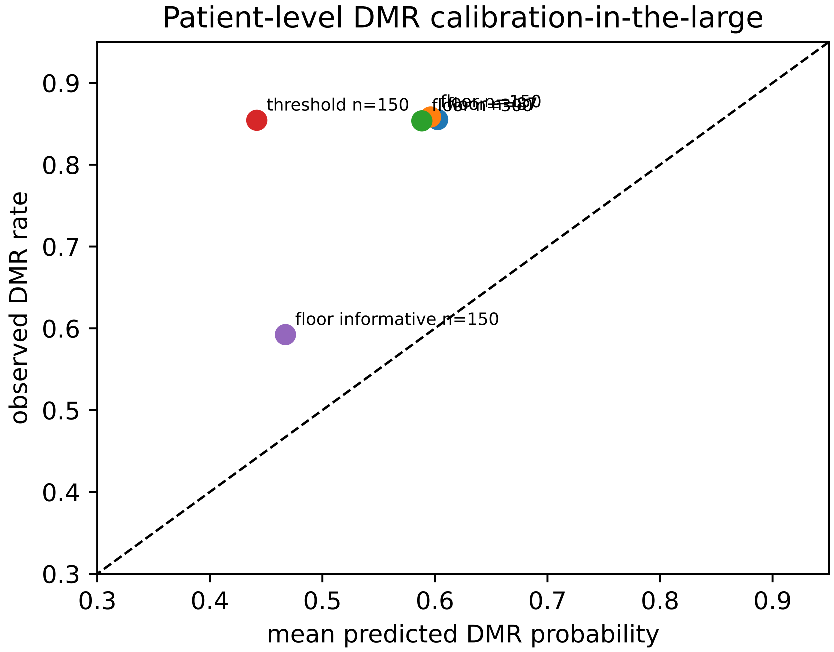

Figure 7: Simulation calibration-in-the-large: mean predicted DMR probability versus observed documented-DMR rate, by estimator, sample size, and monitoring scheme. Points below the diagonal indicate under-prediction.

rate 0.50, relapse among responders 0.26). This check used only 12 replicates at $n = 300$ and is therefore illustrative rather than definitive: with 12 replicates the Monte Carlo error on a coverage estimate near 0.95 is approximately 0.06, and on a bias estimate is the reported MCse (Table 12). Within this limited precision the point estimates were close to their generating values and the threshold width σ_{thr} showed small bias (-0.001), consistent with separating threshold ambiguity from residual observation-level variability; a full-scale study (≥ 200 replicates with Hamiltonian Monte Carlo, below) is required before any claim of reliable recovery or identifiability. Model-implied transition probabilities matched empirical rates for all three transitions (onset 0.503 versus 0.502; durable DMR 0.467 versus 0.494; relapse 0.134 versus 0.129), as illustrated in Figure 8.

Table 12: Multi-state extension: parameter recovery over 12 simulation replicates at $n = 300$. Bias and Monte Carlo standard error (MCse) are on the natural scale; β_{time} and β_{time^2} are the linear and quadratic $\log(1 + t)$ coefficients.

Parameter	Truth	Mean	Bias	MCse
β_0	-2.00	-2.009	-0.009	0.023
β_{time}	-3.60	-3.615	-0.015	0.028
β_{time^2}	1.30	1.280	-0.020	0.028
σ_y	0.80	0.780	-0.020	0.016
τ_{b0}	0.55	0.553	0.003	0.015
τ_{b1}	0.75	0.757	0.007	0.012
σ_{thr}	0.15	0.149	-0.001	0.003

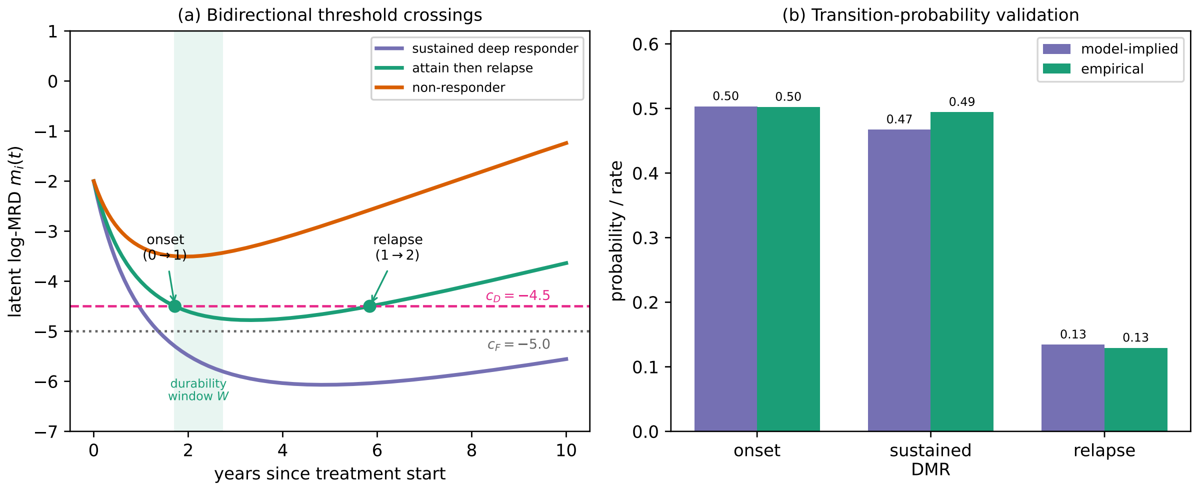

Figure 8: Bidirectional multi-state threshold model. (a) Example latent log-MRD trajectories: a sustained deep responder, a patient who attains then relapses (with onset and relapse crossings of c_D and the durability window w shaded), and a non-responder whose minimum never reaches c_D. (b) Model-implied versus empirical probabilities for the onset, durable-DMR, and relapse transitions.

7 Discussion

7.1 Principal findings

We have developed a Bayesian framework that aligns the statistical model with the structure of routine CML molecular monitoring: repeated MRD measurements, assay-floor reporting, irregular visits, and interval-observed DMR. In the motivating cohort, the fitted joint longitudinal–interval model was computationally stable, calibrated in the large at the interval level, and outperformed Kaplan–Meier, interval-only, landmark, and exact-floor alternatives by Brier score. The comparison against the exact-floor joint model shows that treating deep values as left-censored, rather than as exact -5.0 measurements, measurably improves fit. A simulation study confirms this is a necessity rather than a refinement: ignoring floor censoring roughly halves the recovered random-slope heterogeneity (Section 6). The companion multi-state threshold-crossing model resolved the same endpoint into onset, durable response, and molecular relapse; separating the threshold width from residual observation-level variability improved its identifiability, and a piecewise-quadratic trajectory made the running minimum and maximum closed-form, so the model sampled without divergences and—with a spline trajectory—captured molecular relapse in the cohort.

7.2 Methodological contribution

The methodological contribution is a joint model that couples a left-censored longitudinal submodel for log-MRD with an interval-observed likelihood for first documented DMR through shared random effects, fitted in a fully Bayesian way. Treating assay-floor measurements as left-censored, rather than as exact -5.0 values, is not a refinement but a necessity: in simulation it materially changes the recovered between-patient heterogeneity. The second, and more novel, contribution is the latent threshold-crossing formulation and its multi-state extension. Reading DMR as a first-hitting time of the latent trajectory removes the definitional circularity between the biomarker predictor and the endpoint; separating the threshold width σ_{thr} from the residual observation-level variability σ_y makes both identifiable; and, because the trajectory is piecewise-quadratic, the running minimum and maximum that define onset, durability, and relapse are available in closed form, eliminating the soft-minimum grid and the divergences it induces. A patient-specific spline slope change lets the trajectory rebound and thereby reproduce molecular relapse while preserving the closed form. Together these give a single latent process from which onset, durable response, and relapse are read off as successive threshold crossings.

7.3 Clinical interpretation

The fitted model should be read as a monitoring-support framework, not a treatment-decision rule. The negative association between latent MRD and interval DMR probability is biologically coherent and supports using serial molecular history for response assessment, but it does not validate the model for TKI discontinuation, treatment-free-remission eligibility, or changes in treatment. Dynamic prediction with recalibration improves calibration-in-the-large and provides individualized horizon probabilities, but the current estimates are apparent rather than strictly prospective.

7.4 Informative monitoring

Molecular monitoring in CML is response-adaptive by design—guidelines intensify testing when the transcript rises—so the visit process is potentially informative, and we assessed its impact directly rather than relegating it entirely to the appendix. Under a data-generating scheme in which visit intensity depends on the latent trajectory (Section 6; Supplementary Material), jointly modelling the visit process with the longitudinal and event sub-models left the estimated longitudinal decline (β_{time}) and the threshold-crossing probabilities essentially unchanged relative to the schedule-conditional analysis; the visible effect of informative monitoring was on calibration-in-the-large—the documented-DMR rate fell and the calibration slope rose above one (1.16)—rather than on the structural parameters. Because the additional visit-intensity parameters are only weakly identified in a cohort of this size, and because the structural estimates are robust to including them, we regard conditioning on the observed visit schedule as a reasonable approximation for the present cohort, while noting that a fully joint visit-process analysis is the appropriate route for larger series where the intensity parameters can be estimated precisely.

7.5 Limitations

Several limitations temper these findings. The cohort was modest (87 patients; complete core covariates in 62) and came from a single centre in Kunming, Yunnan Province, China, with no external validation cohort; the relatively young age distribution is typical of Chinese CML series and may limit transportability. The finite-sample interval coverage of the estimator was imperfect at the cohort sizes studied, so we describe the floor-censored estimator as recovering the generating parameters more closely than the exact-floor coding rather than as unbiased. Monitoring was irregular, so patients with longer or denser follow-up had more opportunity to document DMR, and the analysis conditions on the observed visit schedule. The retained data contained no observations in $(-5.0, -4.5]$, limiting the ability to distinguish DMR from CMR descriptively. The dynamic predictions use each patient's full longitudinal record to estimate random effects and are therefore internal (apparent) rather than strictly prospective. Molecular relapse was uncommon (9 confirmed events) and was captured only after enriching the trajectory with a spline slope change, so the relapse estimates are the least certain part of the analysis and warrant larger cohorts. Informative monitoring is not modelled in the primary analyses and is left to future work (Supplementary Material). The interval-observed joint model links the documented-DMR event to the latent trajectory through the association parameter α; the threshold-crossing model removes the residual definitional circularity between predictor and endpoint directly. The models are intended to support monitoring, not to serve as a treatment-decision rule.

7.6 Future work

Several extensions follow naturally. The informative visit process can be modelled jointly with the trajectory as a sensitivity analysis in larger cohorts (Supplementary Material). The multi-state relapse component would benefit from cohorts with more confirmed relapses and from priors elicited for the spline rebound. Most importantly, strict prospective landmark prediction with history-restricted posterior updating, and external or temporal validation, are needed before the recalibrated dynamic probabilities could support clinical decisions.

8 Conclusions

We have presented a Bayesian joint model for CML molecular monitoring in which a single latent log-MRD trajectory serves both the repeated measurements and the event process, and deep molecular response is read directly off that trajectory as a threshold crossing. Reading the endpoint this way removes the definitional circularity that arises when an endpoint defined by a biomarker threshold is predicted from the same biomarker; because the trajectory is piecewise-quadratic, the running extrema that define crossings are closed-form and the model samples without divergences. The conventional interval-censored hazard and a multi-state extension arise as options of the same program, which makes them directly comparable and allows the threshold width and trajectory curvature to be assessed for stability.

Three findings are robust. The biomarker–event association is stable across the censoring point, prior families, and every misspecification examined. Treating assay-floor values as left-censored rather than exact is a necessity rather than a refinement: coding them as exact roughly halves the recovered between-patient heterogeneity. And the single-program formulation reproduces the conventional specification exactly, so the alternatives differ in what they assume, not in how they were fitted. Three claims are correspondingly limited: the threshold width is conditionally rather than universally identified, varying with which transitions are modelled; the trajectory curvature is identified only when multi-state transitions are included, so we do not interpret it as a description of long-run dynamics; and the relapse component rests on few confirmed events and is exploratory. Dynamic predictions are reported as internal accuracy; informative-visit modelling, prospective calibration, and external validation are required before clinical decision use.

Data and Code Availability

The raw clinical molecular-monitoring records cannot be shared publicly because they constitute identifiable patient health information and are subject to institutional and national privacy regulations. To support reproducibility, we provide (i) a fully synthetic dataset of the same structure, simulated from the fitted model's posterior predictive distribution, which reproduces the qualitative features of the analysis without disclosing any patient data, and (ii) the complete, fully commented Stan code for the joint longitudinal–interval model and for the closed-form piecewise-quadratic multi-state threshold-crossing model, together with the R scripts that prepare the data, fit the models, and generate all tables and figures. These materials are included in the Supplementary Material and are maintained in a public repository (URL provided on acceptance; archived at Zenodo with a DOI to be assigned). The synthetic-data workflow allows the full analysis pipeline to be executed end-to-end without access to the confidential clinical data.

Statements and Declarations

Ethical considerations

The study used de-identified, retrospectively collected molecular monitoring data. The relevant Institutional Review Board / Ethics Committee approval or waiver (name, institution, and approval number) is provided on the Title Page and will be stated here in the non-anonymized version.

Consent to participate

As detailed on the Title Page, the requirement for informed consent was addressed under the approving committee's determination for retrospective, de-identified data.

Consent for publication

Not applicable; the manuscript reports de-identified aggregate analyses and contains no individually identifying data.

Declaration of conflicting interests

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

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Data availability

The analysis workflow generated privacy-preserving processed datasets, model code, tables, and figures. Direct identifiers are excluded from all analysis files and retained only in a private linkage file; processed data and code supporting the findings are available from the corresponding author on reasonable request, subject to institutional and ethical approval.

Reporting guidelines

Development and reporting of the prediction model followed the TRIPOD recommendations; the completed checklist is provided as a supplementary file.

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